

FIELD THEORY PART-II

Phase 7: Time-Varying Electromagnetic Fields

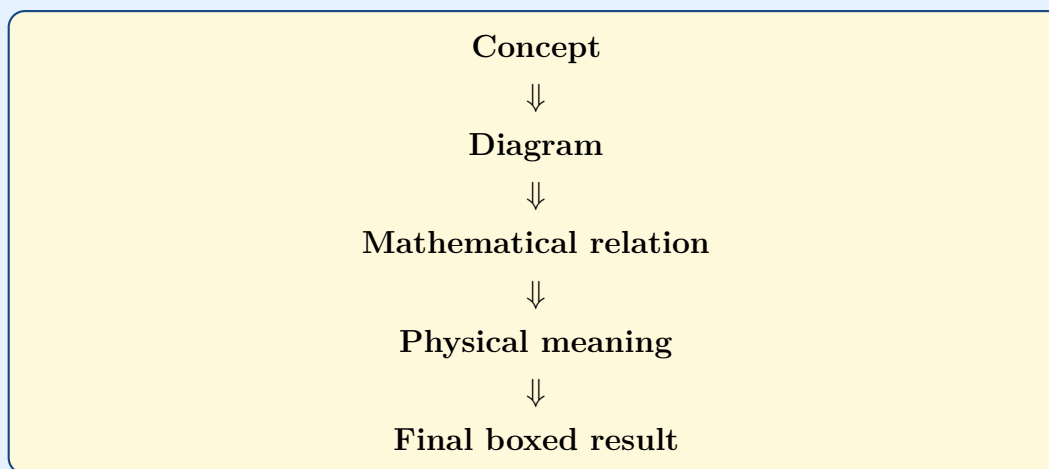
Displacement current, modified Ampere law and Faraday law

Academic Article Style Notes for Semester Preparation

How to Use This Phase

This phase is made for semester preparation from a very basic level. Read the concept first, then understand the diagram, then learn the derivation pattern, and finally practise the exam-style writing block.

The recommended writing order is:



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1 Aim of Phase 7

Main Goal

This phase explains time-varying electromagnetic fields: displacement current, modified Ampere law, Faraday law and continuity equation.

2 Need for Time-Varying Field Theory

Main Idea

In time-varying fields, electric and magnetic fields change with time. Because of this, electric and magnetic fields can produce each other.

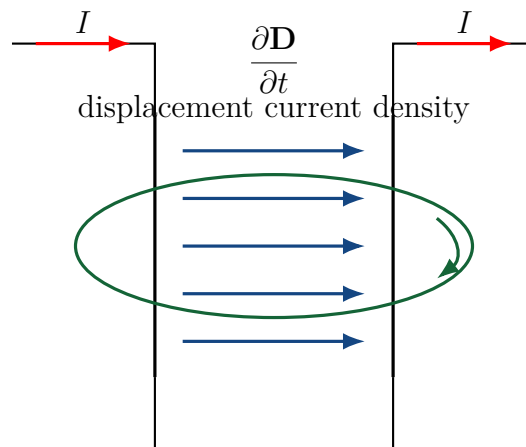
3 Displacement Current

Definition

Displacement current density is:

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$$

It is a current-like effect due to changing electric flux density.



Changing electric field between plates acts like current

Figure 1: Displacement current between capacitor plates.

4 Modified Ampere Law

Complete Law

Old Ampere law:

$$\nabla \times \mathbf{H} = \mathbf{J}.$$

Maxwell added displacement current:

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Integral form:

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \iint_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{S}$$

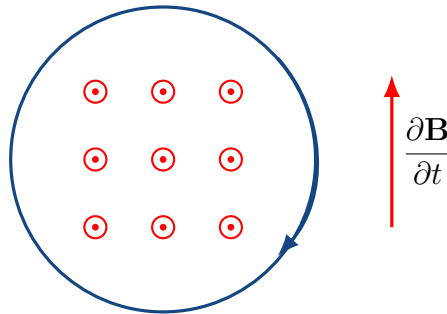
5 Faraday's Law

Basic Meaning

A changing magnetic field produces a circulating electric field.

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Changing magnetic field



Induced circulating electric field $\mathbf{E}$

Figure 2: Faraday law pictorial view.

6 Derivation of Faraday Differential Form

From Integral Form

Faraday law:

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{S}.$$

For fixed surface:

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = - \iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}.$$

Using Stokes' theorem:

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S}.$$

Therefore:

$$\boxed{\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}}$$

7 Continuity Equation

Charge Conservation

$$\boxed{\nabla \cdot \mathbf{J} = - \frac{\partial \rho_v}{\partial t}}$$

This means current cannot disappear or appear without changing charge density.

8 Exam Writing Template

Modified Ampere Law

Write:

1. Old Ampere law.
2. Need for displacement current.
3. Define J_d .
4. Integral form.
5. Stokes theorem to differential form.
6. Capacitor diagram.

Faraday Law

Write:

1. Integral law.
2. Apply Stokes theorem.
3. Compare surface integrals.
4. Explain negative sign as Lenz's law.
5. Draw circular electric field diagram.

9 Final Self-Test for This Phase

Instructions

Answer these without looking at the notes. If you cannot answer a point, revise that section again before moving to the next phase.

1. Can I state the main definition of this phase?
2. Can I draw the main diagram neatly without overlap?
3. Can I explain the physical meaning in two or three lines?
4. Can I write the final formula with correct symbols?
5. Can I connect this topic with at least one past-year question pattern?
6. Can I write a semester-style answer with definition, diagram, derivation and final result?

Completion Rule

This phase is complete only when the concept, diagram, formula and answer-writing pattern are all ready.